



Figure 3: Dockerised versions of Linux distributions are quite popular (Source: Carbon Black)

machine almost fully replicates the operation of a parallel operating system, it requires a proportion of the RAM, storage and processing power when run on a machine. That said, it is one of the most full-fledged representations, causing minimal problems if processing and storage requirements are met.

Typically, installing a virtual machine is simplified by tools like Virtualbox, VMware Player, etc, which only require you to download an ISO image of an OS, as you would need to do for installation on a standard computer. The rest of the process involves just system configuration, and you are good to go!

Advice for those trying out this option would be to ensure the storage and memory limits do not severely restrain the regular operation of your system while trying to spin up a virtual machine.

Cygwin

Cygwin is structured to serve as a set of tools that allows Linux software to be run on Windows as long as it has been recompiled for this purpose. Much of the software available for Linux already fits this bill, but the subtle difference that needs to be highlighted is that it does not offer a Linux environment but rather a Linux-like environment for Windows. For those users who miss the Linux command line and relevant programs, this might be a viable alternative to try and replicate the feel of the Linux environment for Windows.

Note that Cygwin has an ecosystem supported by a large set of packages designed to enhance the command-line experience for Windows, but only the base packages are installed by default. Users are free to choose what additional packages should be installed within their systems.

Docker and Linux containers for Windows

The rise of Docker seriously changed the rules of the game for running most software tools in the cloud. Windows Server, one of the industry standards and a widely used operating system, released new features supporting containerised applications in coordination with Docker. In fact, the release announcement at DockerCon was extremely well received by the understandably-relieved audience comprising many developers involved in working with containerised applications. Sure, this does not technically mean Linux is running on a system; but there exist containerised images that, by this definition, can be run on Windows machines. In fact, containerised images of most Linux distributions are now widely available—albeit some comprise barebones command-line functionality.

Developers were eager to run any container, Windows or Linux, on the same machine. This feature, as the Docker Team announced, uses Hyper-V isolation to run a kernel with just enough of an OS to support running containers. The feature is not only available on Windows Server but also on Windows 10 for those interested in trying it out.

Windows Subsystem for Linux (WSL)

The best way to run Linux on Windows 10 may just be Microsoft's introduction of Bash and consequently the option to 'install' Ubuntu and some other Linux distributions via the Microsoft Store. The reasoning behind this effort was quite simple – developers using Windows also need to run Linux, and Windows not only made itself a more attractive platform for software and Web developers but also made good on its commitment to open source. Earlier, testing Web apps and software required a convoluted chain of getting tools to work in sync in order to simulate and run the tests for Linux tools on Windows machines. The introduction of WSL has dramatically reduced this workflow. It has simply added enough support to convince developers that life will be made easier eventually, and they may not always have to switch around different environments while building different tools.

The configuration of the WSL environment involves running a PowerShell command to set permissions on an optional feature, followed by a quick search and installation of the desired Linux distribution from the Microsoft Store. You can choose from Ubuntu, openSUSE, Kali Linux and Debian. It's really that simple, so go check it out! **END** 

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